stm32_k230二维码识别

stm32_k230二维码识别
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k230和stm32通信

1. 实验前提

本教程使用的是STM32F103C8T6开发板，对应例程路径为【14.export\STM32-K230\03_STM32_k230_qrcode】。

k230要运行【14.export\CanmvIDE-K230\03.find_qrcodes.py】程序才能开始实验，建议下载为离线运行程序。

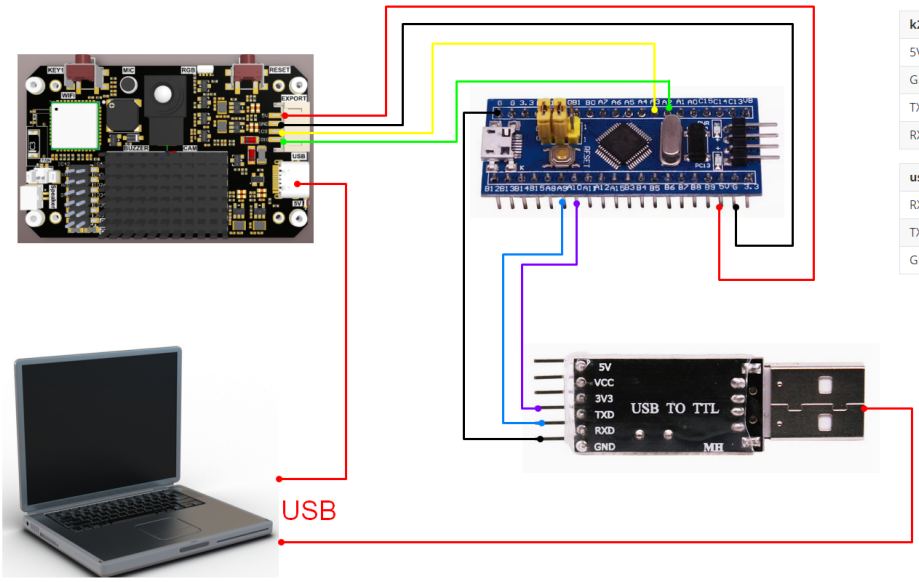
需要用到的物品：

Windows系统电脑
STM32F103C8T6开发板
usb转ttl模块
K230视觉模块(包含烧录好镜像的TF卡)
type-C数据线
连接线（杜邦线）

2. 实验接线

k230视觉模块	**STM32**
5V	VCC
GND	GND
TXD(IO9)	PA3
RXD(IO10)	PA2

usb转ttl模块	**STM32**
RXD	PA9
TXD	PA10
GND	GND



k230视觉模块	STM32
5V	VCC
GND	GND
TXD(IO9)	PA3
RXD(IO10)	PA2

usb转ttl模块	STM32
RXD	PA9
TXD	PA10
GND	GND

3. 主要代码讲解

```
void Pto_Data_Parse(uint8_t *data_buf, uint8_t num)
{
    uint8_t pto_head = data_buf[0];
    uint8_t pto_tail = data_buf[num-1];
    if (!(pto_head == PTO_HEAD && pto_tail == PTO_TAIL))
    {
        printf("pto error:pto_head=0x%02x , pto_tail=0x%02x\n", pto_head,
pto_tail);
        return;
    }
    uint8_t data_index = 1;
    uint8_t field_index[PTO_BUF_LEN_MAX] = {0};
    int i = 0;
    int values[PTO_BUF_LEN_MAX] = {0};
    char msg[PTO_BUF_LEN_MAX] = {0};
    for (i = 1; i < num-1; i++)
    {
        if (data_buf[i] == ',')
        {
            data_buf[i] = 0;
            field_index[data_index] = i;
            data_index++;
        }
    }

    for (i = 0; i < data_index; i++)
    {
        if (i == 6)
        {
            memcpy(msg, (char*)data_buf+field_index[i]+1, values[0]-
field_index[i]-2);
        }
        else
        {
            values[i] = Pto_Char_To_Int((char*)data_buf+field_index[i]+1);
        }
```

```
    }

    uint8_t pto_len = values[0];

    if (pto_len != num)
    {
        printf("pto_len error:%d , data_len:%d\n", pto_len, num);
        return;
    }
    uint8_t pto_id = values[1];
    if (pto_id != PTO_FUNC_ID)
    {
        printf("pto_id error:%d, func_id:%d\n", pto_id, PTO_FUNC_ID);
        return;
    }
    int x = values[2];
    int y = values[3];
    int w = values[4];
    int h = values[5];
    printf("qrcode:x:%d, y:%d, w:%d, h:%d, msg:'%s'\n", x, y, w, h, msg);
}
```

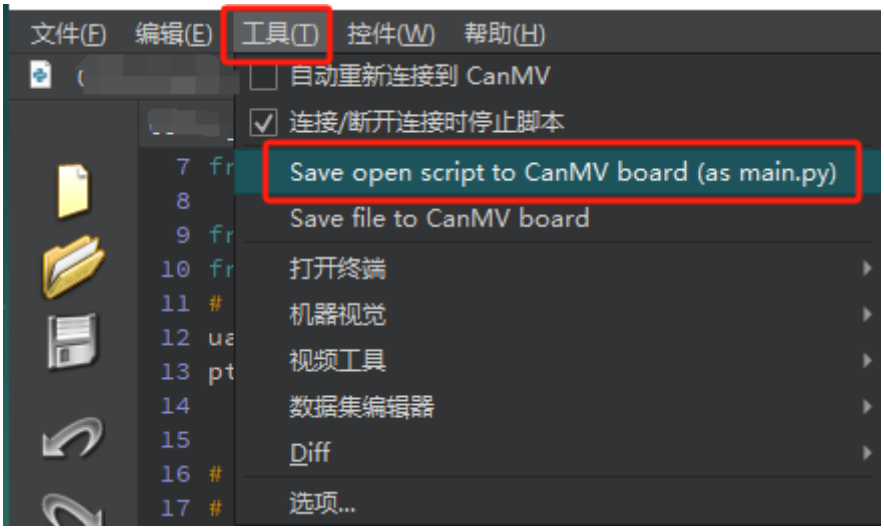
以上函数为解析K230数据的函数，只有符合特定协议才可以解析出对应的数据。

其中

- x：是识别出来框左上角的横坐标
- y：是识别出来框左上角的纵坐标
- w：是识别出来框的宽度
- h：是识别出来框的长度
- msg：是二维码的内容

4. 实验现象

1. 连接好线后，k230视觉模块脱机运行
   K230连接Canmv IDE后，打开对应程序，点击工具栏的 【Save open script to CanMV board (as main.py)】，然后重启K230。



2. STM32烧录例程生成的hex文件。



3. 串口助手设置成如图的界面



4. 当K230摄像头画面识别到二维码时，串口助手就会打印出k230传输给stm32的信息。

其中

- x：是识别出来框左上角的横坐标
- y：是识别出来框左上角的纵坐标
- w：是识别出来框的宽度
- h：是识别出来框的长度
- msg：是二维码的内容

如下图所示

```
qrcode:x:235, y:80, w:154, h:208, msg:'goahead'
qrcode:x:236, y:79, w:155, h:203, msg:'goahead'
qrcode:x:237, y:75, w:154, h:207, msg:'goahead'
qrcode:x:238, y:74, w:156, h:206, msg:'goahead'
qrcode:x:240, y:72, w:155, h:209, msg:'goahead'
qrcode:x:242, y:67, w:155, h:202, msg:'goahead'
qrcode:x:244, y:62, w:155, h:209, msg:'goahead'
qrcode:x:246, y:57, w:155, h:206, msg:'goahead'
qrcode:x:247, y:52, w:155, h:204, msg:'goahead'
qrcode:x:251, y:42, w:157, h:206, msg:'goahead'
qrcode:x:254, y:35, w:155, h:206, msg:'goahead'
qrcode:x:257, y:28, w:157, h:204, msg:'goahead'
qrcode:x:261, y:12, w:157, h:208, msg:'goahead'
```